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Intrinsic Risk Factors For the Development of Anterior Knee Pain in an Athletic Population

A Two-Year Prospective Study*

Erik Witvrouw,†‡§ PT, PhD, Roeland Lysens,‡ MD, PhD, Johan Bellemans,‡ MD, PhD, Dirk Cambier,† PT, PhD, and Guy Vanderstraeten,† MD, PhD

From the †Department of Physical Therapy, Faculty of Medicine, University of Gent, Belgium, and ‡Faculty of Physical Therapy & Rehabilitation Sciences, Catholic University of Leuven, Belgium

ABSTRACT

Many variables have retrospectively been associated with the presence of anterior knee pain. Very few prospective data exist, however, to determine which of these variables will lead to the development of anterior knee pain. It was our purpose in this study to determine the intrinsic risk factors for the development of anterior knee pain in an athletic population over a 2-year period. Before the start of training, 282 male and female students enrolled in physical education classes were evaluated for anthropometric variables, motor performance, general joint laxity, lower leg alignment characteristics, muscle length and strength, static and dynamic patellofemoral characteristics, and psychological parameters. During this 2-year follow-up study, 24 of the 282 students developed patellofemoral pain. Statistical analyses revealed a significant difference between those subjects who developed patellofemoral pain and those who did not concerning quadriceps and gastrocnemius muscle flexibility, explosive strength, thumb-forearm mobility, reflex response time of the vastus medialis obliquus and vastus lateralis muscles, and the psychological parameter of seeking social support. However, only a shortened quadriceps muscle, an altered vastus medialis obliquus muscle reflex response time, a decreased explosive strength, and a

hypermobile patella had a significant correlation with the incidence of patellofemoral pain. We concluded that the latter four parameters play a dominant role in the genesis of anterior knee pain and we therefore deem them to be risk factors for this syndrome.

Patella disorders are the most common knee complaints seen in adolescents and young adults.^{4,5,8,26} Research has shown an incidence as high as one in four, and even higher, among athletes.⁹ Despite this high incidence, the exact cause of these disorders remains enigmatic. Many possible mechanisms have been described by which patellofemoral pain might be initiated.^{16,26,46}

The major complaint of patients with patellofemoral pain syndrome is retropatellar pain during activities such as running, squatting, going up and down stairs, cycling, and jumping. Once started, the patellofemoral pain syndrome frequently becomes a chronic problem, forcing the patient to stop sports and other similar activities.

Various authors have attributed this pain to intrinsic and extrinsic risk factors.^{14,37,42} Extrinsic risk factors are related to factors outside the human body, such as the type of sports activity, the manner in which sport is practiced, the environmental conditions, and the equipment used. Intrinsic risk factors relate more to individual physical characteristics and psychological traits. General agreement exists on the classification of the risk factors of patellofemoral pain syndrome into these two categories.

Some authors have undertaken retrospective studies in which they examined these proposed intrinsic risk factors.^{2,15,32,41,51,55,56} However, only one study has been undertaken to scientifically determine these intrinsic risk factors on a prospective basis.⁴²

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§ Address correspondence and reprint requests to Erik Witvrouw, PT, PhD, Department of Physical Therapy (MOREKI), Faculty of Medicine, University of Gent, De Pintelaan 185, 9000 Gent, Belgium.

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The purpose of this study, therefore, was to analyze, with a prospective study design, which of the presumed risk factors play a significant role in the development of the patellofemoral pain syndrome. Determination of the intrinsic risk factors of anterior knee pain in an athletic population over 2 years is the first step in injury prevention and treatment.

MATERIALS AND METHODS

A 2-year prospective study was undertaken of 480 students taking physical education classes who had no history of knee complaints or surgery. All participants gave informed consent and knew the goals of the study. During this 2-year study, 198 subjects stopped the classes for various reasons, but not because of knee problems. The remaining 282 students (151 boys and 131 girls) were used for statistical analyses. The mean age of these students was 18.6 years (range, 17 to 21). The 282 students all followed the same sports program, which consisted of 12 to 14 hours of various sports a week. This program was composed of swimming (2 hours/week), track and field (2 hours/week), gymnastics (2 hours/week), soccer (1 or 2 hours/week), basketball (1 or 2 hours/week), volleyball (1 hour/week), jazz dance (1 hour/week), handball (1 hour/week), and judo (1 hour/week).

Before the program commenced, all students had screening that included a survey of their participation in sports activities other than those in the physical education classes. Before the beginning of the study, all students' knees were examined for signs of ligamentous instability, meniscus abnormalities, effusion, and tenderness. The same doctor and orthopaedic surgeon reviewed the students every 3 months and questioned them about pain in the knee.

To be considered a patellofemoral pain patient, subjects had to have a characteristic history and symptoms of patellofemoral pain syndrome for more than 6 weeks (that is, retropatellar pain during physical activities such as jumping, running, squatting, and going up or down stairs); exhibit two of the following clinical criteria on assessment²⁶: pain on direct compression of the patella against the femoral condyles with the knee in full extension, tenderness of the posterior surface of the patella on palpation, pain on resisted knee extension, and pain with isometric quadriceps muscle contraction against suprapatellar resistance with the knee in 15° of flexion; and negative findings in the examination of knee ligaments, menisci, bursae, synovial plicae, Hoffa's fat pad, iliotibial band, and the hamstring, quadriceps, and patellar tendons and their insertions. Previous studies indicate that these criteria are sensitive and specific for diagnosing patellofemoral pain.^{7,28} Indeed, DeHaven et al.⁷ found tenderness of the posterior patellar facet in 87 of 100 patients with patellofemoral pain syndrome. They identified this as the most frequent physical finding.

All students were also evaluated for anthropometric variables, physical fitness, general joint laxity, lower leg alignment characteristics, muscle length and strength,

static and dynamic patellofemoral characteristics, and psychological parameters.

Anthropometric Evaluation

Anthropometric measurements included height, weight, body fat percentage using the skin caliper method,³⁵ and body composition using the method of Heath and Carter.²⁰

Physical Fitness and Cardiorespiratory Exercise Capacity

Physical fitness was assessed using the Eurofit test.^{44,59} This test examines global balance by means of a flamingo balance test, explosive jump ability by a vertical jump test, the functional strength by a bent arm hang, sprinting capacity by a shuttle run, speed of limb movement by plate tapping, static hand strength by an arm pull, abdominal strength by a sit-up test, and flexibility by a sit and reach test.

Cardiorespiratory exercise capacity was measured by a progressive submaximal ergometer test following the Hollmann protocol.²¹

General Joint Laxity

Each subject's general joint laxity was assessed by measuring apposition of the thumb to the forearm; elbow, knee, shoulder, and little finger hyperextension; and mediolateral mobility of the patella.^{36,39} The latter parameter was measured with the subject's quadriceps muscle at rest and the knees in extension. The examiner applied a maximal manual force medially or laterally to the patella. The displacement was recorded in millimeters in the coronal plane with a measuring rod. After the force was applied, the patella returned passively to its resting position. A gentle posteriorly directed force was applied and released, allowing the patella to return to its original resting position, and the test was repeated. Three tests were performed in each direction, and the average displacements were used in data analysis. An earlier study identified this measurement as reliable.³⁶ To obtain the total patellar mobility, the medial displacement was added to the lateral displacement.

Lower Leg Alignment

Lower leg alignment characteristics were obtained by clinically measuring the leg length difference, the foot alignment, and the muscle length.

The leg-length difference was evaluated by gauging the distance between the anterior superior iliac spine and the medial malleolus of both legs with the subject in the supine position.^{28,36}

Examination of foot architecture included an evaluation of the foot arch and measurement of the lower leg-heel alignment and heel-forefoot alignment. The foot arch was evaluated while the subject was standing on a podograph. The lower leg-heel and heel-forefoot alignment were evaluated using the method described by Hyvarinen and Karhu,²⁴ which has been found to be valid and reliable.¹²

The tightness of the hamstring, quadriceps femoris, and gastrocnemius muscles was measured with a goniometer. Hamstring muscle tightness was tested with the subject in a supine position. The examiner lifted one of the straightened limbs and measured the angle at the hip.¹⁸ The quadriceps femoris muscle was tested with the subject in a prone position. The knee was maximally flexed while the foot on the noninvolved side was placed on the floor in a 90° of hip flexion position. In this position the angle of the knee was measured.¹⁸ The tightness of the gastrocnemius muscle was obtained by having the subject lean on a solid support 0.60 meters (2 feet) away with the tested leg behind the opposite leg and keeping the knee of the tested leg extended. The subjects were instructed to maximally flex their tested ankle while keeping their heel to the floor.

Static Patellofemoral Alignment Characteristics

To obtain some information on static patellofemoral alignment, we measured the Q angle using a goniometer and anatomically placed lines with the subject in a supine position with a relaxed quadriceps muscle.²²

In addition, a clinical genu varum/valgum evaluation was performed as follows: The student was in a supine position with the quadriceps muscle relaxed. Depending on the alignment of the knees, either the medial malleoli or the knees touched. For students in whom the malleoli touched (genu varum), the distance between the medial condyles of the tibias was measured to the nearest centimeter. If the knees touched (genu valgum), the distance between the medial malleoli was measured to the nearest centimeter. If the distances between the medial condyles of the tibias and the medial malleoli were both less than 1 cm, the knees were evaluated as neither varus or valgus, thus normal.

Dynamic Patellofemoral Alignment Characteristics

Isokinetic strength of the quadriceps and hamstring muscles was evaluated on the Cybex 350 dynamometer (Cybex, Inc., Ronkonkoma, New York) as a dynamic patellofemoral characteristic. Concentric knee extensor and flexor peak torques were measured at three speeds: 60, 180, and 240 deg/sec. Each subject was positioned according to the Cybex testing manual (*Cybex I: Isolated Joint Testing and Exercise*. Ronkonkoma, NY, Cybex Inc., 1983).

To determine the response times of the vastus medialis obliquus and vastus lateralis muscles, the Vickers MS92 electromyography unit (Vickers Medical, Old Woking, Surrey, United Kingdom) was used. This unit contains a 20- to 10,000-Hz filter and has a sensitivity of 100 μ V. Each subject was placed with the hips and knees at a 90° angle, with the lower leg hanging over the edge of the table. All tests were performed by the same examiner.

After preparing the skin, the examiner used VELCRO straps (VELCRO USA, Manchester, New Hampshire) to attach a plastic bar containing an active and a reference rectangular surface bar electrode, consisting of two 20 $\times$ 7 mm saline-soaked felt pad electrodes with silver/silver chloride contacts fixed in a plastic mount, to the lower part

of the muscle bellies of the vastus medialis obliquus and vastus lateralis. The electrode placement was standardized by placing the electrodes at this point of the muscle belly, where an initial negative deflection was detected (motor point determination technique).^{1,6,31}

The patellar tendon was stimulated by eliciting the knee jerk reflex with a special reflex hammer, modified for EMG use, that was connected to the electromyography unit. This patella hammer has a quarter-inch jack plug and functions by making a connection to a nonisolated ground when struck. It needs to have a pull-up resistor to 5 V. It produces multiple contact closures lasting 150 ms, the first of which has a variable delay of 3 to 7 ms, depending on speed of strike. A tap on the patellar tendon produced an electric signal, which functioned as an external trigger and started the sweep on the screen (sweep of 50 ms). This tap force needed to be of sufficient intensity to provoke a clinically visible patellar tendon reflex, which resulted in a well-defined motor response on the screen. This distinct tendon reflex always resulted in a motoric answer of at least 200 to 300 μ V. The time from the tendon tap to the onset of the EMG signal from the muscle contraction, where the baseline is at the left, was measured as the reflex response time (in milliseconds).

The testing was repeated 10 times on the vastus medialis obliquus and vastus lateralis muscles for each knee, on each subject, with an interval of 20 seconds. All subjects were tested at both knees. The shortest reflex response time was used for data analysis. We did not observe any habituation effect of the 10 trials in our subjects. The 10 trials were found to be of equal extent, with a difference of a few tenths of a millisecond in each knee on each subject.

Psychological Parameters

After completion of the physical evaluation, psychological characteristics were evaluated. By using the Utrecht Coping List⁴⁹ and the Amsterdam Biographic Questionnaire,⁶² we were able to get an insight into the use of coping mechanisms and get an idea about the intro/extroversion, neurotic, and psychosomatic lability of the students. Both psychological scales have been found to be very reliable for testing adolescents.^{45,54,61}

Statistical Procedures

A variety of statistical techniques were used to analyze the data. Means and standard deviations were calculated for the characteristics. Univariate analyses (independent *t*-tests) were performed to compare the continuous variables between the students who did and did not have patellofemoral pain when there was a normal distribution. A nonparametric Wilcoxon's test was performed when the distribution did not meet the criterion of normality. A chi-square or Fisher's exact test was used to compare the two groups of students for the discrete variables. Significance was accepted at the 0.05 level. Since there was an unequal sex distribution between groups, and some of the

measured parameters were significantly influenced by sex, we transformed the data so they became sex-independent.

If one wants to study the relationship between a binary response variable (in our case, patellofemoral pain syndrome) and one or more explanatory variables (evaluated parameters), the logistic regression model has become, in many fields, the standard method of analysis in this situation.²³ Analysis of variance can also be used to investigate this relationship. However, in the analysis of variance, the explanatory variables are not treated as quantitative (as in the logistic regression model), but only as qualitative, and relations among the explanatory variables are not taken into account.⁵⁰ Therefore, to identify the intrinsic risk factors in this study, a multivariate analysis was performed with the use of stepwise logistic regression. Because of the different techniques used in the analysis of variance and the logistic regression, it is not impossible (although rather rare) that one variable is significant with one method but not with the other (in this study, this was the case for patellar mobility). If one is interested in predicting the probability of the binary variable on the basis of one or more explanatory variables, undoubtedly the results of the logistic regression analysis should be used for interpretation.^{23, 50, 63}

Power analysis revealed that the statistical power for the univariate and multivariate analysis exceeded 80%.

RESULTS

During this 2-year follow-up period, the patellofemoral pain syndrome was diagnosed in 24 (9%) of the 282 students. Of the 151 boys, 11 (7%) developed patellofemoral pain syndrome. Of the 131 girls, 13 (10%) developed patellofemoral pain syndrome. Eleven subjects had bilateral patellofemoral pain. Of the 13 subjects with unilateral pain, 8 suffered from patellofemoral pain in their dominant leg.

No significant difference was found in the amount of external sports activity between the students who did and did not have patellofemoral pain. Students with patellofemoral pain participated in sports outside their education program for 2.8 hours a week, while the control students participated in 2.5 hours a week of external sports activities ($P = 0.77$).

Concerning the context of the external sports participation, 84% of the students without patellofemoral pain (217

of 258) participated in competitive sports. Since 67% of the students with patellofemoral pain (16 of 24) participated in competitive sports, statistical analysis showed no significant difference in context of sports participation between the two groups ($P = 0.18$).

Regarding the possible differences in the type of sports activity between the two groups, it must be recognized that because of the wide variety of sports activities and the small number of students with patellofemoral pain syndrome in this study, it was impossible to make a statistical comparison.

Anthropometric Evaluation

Univariate analysis did not reveal any significant differences between the students who did and did not have patellofemoral pain for any of the measured anthropometric variables ($P > 0.05$) (Table 1).

Physical Fitness and Cardiorespiratory Exercise Capacity

The findings of the physical fitness and cardiorespiratory exercise capacity are reported in Table 2. These results indicated a significant difference between the groups only in vertical jump performance. The group with patellofemoral pain showed a statistically significantly lower explosive strength capacity compared with the control subjects ($P = 0.01$).

General Joint Laxity

A univariate comparison of values of the different general joint laxity tests was made between the two groups (Table 3). Only in thumb-forearm mobility did the group with patellofemoral pain show a significantly greater range of motion compared with the control group ($P = 0.01$). Of note is the fact that the values for the medial, lateral, and total patellar mobility were all greater in the patellofemoral pain group, although these differences were not statistically significant.

Lower Leg Alignment

Results of the statistical analysis indicate that there was no significant difference in leg-length dissimilarity between the students with patellofemoral pain and those

TABLE 1
Univariate Analysis of the Different Anthropometric Variables in Subjects with Patellofemoral Pain Syndrome (PFPS) and Subjects with No Pain (Control)

Variable	PFPS (N = 24)		Control (N = 258)		P-value
	Mean	SD	Mean	SD	
Height (cm)	179.3	5.38	180.16	6.25	0.59
Weight (kg)	68.14	5.59	69.96	6.85	0.21
Endomorphism ^a	2.32	0.71	2.49	0.86	0.43
Ectomorphism ^a	3.58	0.75	3.39	0.84	0.36
Mesomorphism ^a	3.70	0.68	3.96	0.75	0.16
Fat percentage	11.51	2.80	12.13	3.30	0.44
Ponderal index	43.84	1.03	43.61	1.10	0.39

^a Rating based on the method of Heath and Carter.²⁰

TABLE 2

Univariate Analysis of the Different Physical Fitness Tests of the Eurofit Test and the Cardiorespiratory Exercise Capacity of Subjects with Patellofemoral Pain Syndrome (PFPS) and Subjects with No Pain (Controls)

Test	PFPS (N = 24)		Control (N = 258)		P-value
	Mean	SD	Mean	SD	
Flamingo balance (sec)	8	3.22	6.98	3.77	0.21
Vertical jump (cm)	52.63	3.63	56.02	6.28	0.01 ^a
Standing broad jump (cm)	240.96	13.77	243.89	16.35	0.41
Bent arm hang (sec)	44.88	16.95	43.24	13.63	0.60
Shuttle run (sec)	20.15	1.12	20.22	1.21	0.80
Plate tapping (no. repetitions)	92.96	9.88	93.48	8.65	0.79
Arm pull (no. repetitions)	69.42	7.52	72.69	10.91	0.17
Leg lifts (no. repetitions)	17.74	1.51	17.99	1.67	0.49
Sit and reach (no. repetitions)	28.24	6.90	29.86	6.31	0.25
Sit ups (no. repetitions)	26.56	3.43	27.18	3.40	0.41
Maximal O ₂ -uptake (ml/kg/min)	55.09	10.14	54.99	9.81	0.96

^a Significant difference between groups ($P < 0.05$).

TABLE 3

Univariate Analysis of the Different General Joint Laxity Tests in Subjects with Patellofemoral Pain Syndrome (PFPS) and Subjects with No Pain (Control)

Test	PFPS (N = 24)		Control (N = 258)		P-value
	Mean	SD	Mean	SD	
Extension little finger (deg)	88.71	29.53	84.86	28.17	0.58
Mobility shoulders (cm)	109.83	21.93	100.92	21.22	0.06
Extension elbow (deg)	180.81	4.39	182.85	4.12	0.41
Thumb-forearm (deg)	105.06	13.40	86.65	13.51	0.01 ^a
Extension knee (deg)	185.21	6.32	181.53	5.62	0.37
Medial patellar mobility (cm)	1.92	0.35	1.82	0.38	0.20
Lateral patellar mobility (cm)	0.92	0.24	0.80	0.30	0.06
Total patellar mobility (cm)	2.85	0.51	2.62	0.58	0.06

^a Significant difference between groups ($P < 0.05$).

without ($P = 0.355$). When we compared the descriptive data for foot alignment, statistical analysis did not reveal a significant difference in the division of foot type between the students who did and did not have patellofemoral pain ($P = 0.54$).

Means and standard deviations of the muscle flexibility of the quadriceps, hamstring, and gastrocnemius muscles are reported in Table 4. There was a significant difference noticed in flexibility of the quadriceps and gastrocnemius muscles between the group with patellofemoral pain and the control subjects. For both muscles the patellofemoral pain subjects showed significantly lower values.

Static Patellofemoral Alignment Characteristics

Statistical analysis of the Q angle revealed no significant differences between the students with patellofemoral pain

and those without ($P = 0.394$). The observed values in this study (mean, 11.45 ± 6.23 for the patellofemoral pain group; and mean, 13.01 ± 7.66 for the control group) corresponded well with the normal values reported in the literature.^{5,26} Comparisons between groups concerning the spread of genu varum/valgum did not attain significance ($P = 0.96$).

Dynamic Patellofemoral Alignment Characteristics

Figure 1 shows the results for both groups of the isokinetic measurement of quadriceps and hamstring muscle strength at 60, 180, and 240 deg/sec. Differences between the subjects with patellofemoral pain and those without were not statistically significant ($P > 0.05$). Despite this lack of significant difference between groups, the subjects with patellofemoral pain demonstrated a lower value at

TABLE 4

Univariate Analysis of the Flexibility of the Different Measured Muscles in Subjects with Patellofemoral Pain Syndrome (PFPS) and Subjects with No Pain (Control)

Muscle	PFPS (N = 24)		Control (N = 258)		P-value
	Mean	SD	Mean	SD	
Hamstring (deg)	90.78	20.06	93.60	16.47	0.442
Quadriceps (deg)	124.62	12.46	132.21	16.39	0.028 ^a
Gastrocnemius (deg)	32.12	5.35	35.22	6.59	0.038 ^a

^a Significant difference between groups ($P < 0.05$).

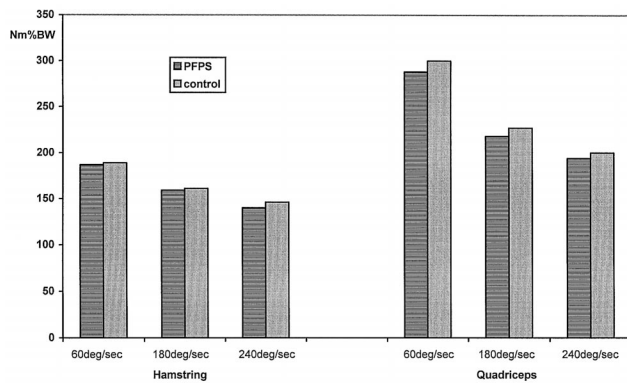


Figure 1. The mean isokinetic strength values of the quadriceps and hamstring muscles (in newton-meters per percent body weight) in subjects with patellofemoral pain syndrome (PFPS) ($N = 24$) and subjects with no pain (Control) ($N = 258$).

each test velocity for the quadriceps muscle compared with the control group. Regarding the hamstring muscles, only a minor difference in strength could be observed between groups, and this difference was seen only at the highest test velocity.

Table 5 illustrates the difference in reflex response times of the vastus medialis obliquus and vastus lateralis muscles in the group with patellofemoral pain and those without. Significant differences were seen between groups for the reflex response time of both muscles. For both muscles the group with patellofemoral pain showed a significantly faster reflex response time compared with the control group. When subtracting the reflex response time

of the vastus lateralis muscle from the reflex response time of the vastus medialis obliquus muscle, statistical analysis failed to reveal a significant difference between groups ($P = 0.26$). Although no significant difference was found for this parameter, we observed an almost equal firing sequence of the two muscles in the group with patellofemoral pain, whereas we noticed a distinctly earlier reflex response time of the vastus medialis obliquus compared with the vastus lateralis in the group without patellofemoral pain.

Psychological Parameters

Table 6 shows the results of the various measured psychological parameters. A significantly lower value for the parameter "looking for social support" was observed for the group with patellofemoral pain compared with the control group ($P = 0.03$), which indicates that the patellofemoral pain subjects looked for social support significantly less than those without pain. In addition, the patellofemoral pain group scored almost significantly ($P = 0.05$) lower for palliative reaction. This denotes that these students divert their attention less from a problem than the control subjects do, and that they find it harder to relax when they are confronted with a puzzle.

Intrinsic Risk Factors

Although several of the measured parameters did yield statistically significant differences between groups by univariate statistics, only a logistic regression technique can

TABLE 5

Univariate Analysis of the Reflex Response Time (in Milliseconds) of the Vastus Medialis Obliquus (VMO) and Vastus Lateralis (VL) Muscles in Subjects with Patellofemoral Pain Syndrome (PFPS) and Subjects with No Pain (Controls)

Muscle	PFPS ($N = 24$)		Control ($N = 258$)		<i>P</i> -value
	Mean	SD	Mean	SD	
VMO	17.70	2.66	18.81	2.16	0.02 ^a
VL	17.76	2.71	19.12	2.07	0.006 ^a
VMO-VL	-0.06	1.37	-0.31	1.39	0.26

^a Significant difference between groups ($P < 0.05$).

TABLE 6

Univariate Analysis of the Measured Psychological Parameters of Subjects with Patellofemoral Pain Syndrome (PFPS) and Subjects with No Pain (Controls)

Parameter	PFPS ($N = 24$)		Control ($N = 258$)		<i>P</i> -value
	Mean	SD	Mean	SD	
Neurotic unstable	58.96	15.00	60.44	24.03	0.77
Psychosomatic unstable	17.65	3.39	17.85	4.99	0.85
Extroversion	62.17	13.30	63.49	14.69	0.68
Active behavior	17.33	3.03	18.49	4.35	0.62
Palliative reaction	16.96	2.97	18.32	3.39	0.05 ^a
Avoiding behavior	15.75	3.00	15.57	2.98	0.78
Looking for social support	13.00	4.34	14.78	3.91	0.03 ^a
Passive reaction pattern	11.00	2.59	11.94	3.27	0.17
Expression of emotions	7.04	1.60	6.97	1.51	0.83
Reassuring thoughts	11.92	1.91	12.38	2.42	0.37

^a Significant difference between groups ($P < 0.05$).

indicate which variables contribute significantly to the development of the patellofemoral pain syndrome.

Among all the variables that were assessed before the start of the physical education program, the stepwise logistic regression identified only four parameters as intrinsic risk factors for patellofemoral pain syndrome. With the use of this technique, only a decreased muscular flexibility of the quadriceps muscle ($P = 0.02$), a shortened reflex response time of the vastus medialis obliquus muscle ($P = 0.005$), a reduction of the vertical jump performance ($P = 0.02$), and an increase in medial patellar mobility ($P = 0.026$) were significantly associated with the occurrence of the patellofemoral pain syndrome.

DISCUSSION

Although patellofemoral pain is frequently encountered in the sports injury clinic, the causes of this syndrome remain enigmatic. In the orthopaedic literature, a wide variety of causes of anterior knee pain have been identified, ranging from osteoarthritis to dislocations, and chondromalacia to synovitis.^{16,26,46} Distinguishing among these possible causes can be challenging, even for experienced clinicians. One must be aware of the fact that today the impact of failure to differentiate between these various causes of anterior knee pain is rather large. Even with the best clinical criteria we cannot always discern patellofemoral pain from synovitis. If the language we use to identify this pain syndrome reflects our understanding of it, it seems clear that we have only a limited understanding of the processes at work. It is our opinion that the term patellofemoral pain syndrome is still a "wastebasket" term that may be used to classify different entities. This may be an important reason for the conflicting results among different studies. We tried to improve our insight into a few of these possible causes of patellofemoral pain syndrome.

Many authors cite both extrinsic and intrinsic parameters as being etiologic factors for patellofemoral pain syndrome.^{14,27,37,42} The association between clinical overloading of the joint (extrinsic parameters) and patellofemoral pain is well known.^{14,42} However, the relationship between intrinsic parameters and patellofemoral pain syndrome is still obscure. Prospective studies that investigate the impact of the assumed intrinsic factors on the development of the patellofemoral pain are especially scarce. Kujala et al.³³ studied factors predisposing army conscripts to exertion-related injuries involving the knee, but they did not look specifically at patellofemoral pain syndrome. Milgrom et al.⁴² are, to our knowledge, the only authors who have undertaken a prospective study of patellofemoral pain syndrome. These authors prospectively examined 390 male infantry recruits during 14 weeks of basic training. During this period, patellofemoral pain related to overactivity was diagnosed in 77 knees of 60 recruits.

In our present study we evaluated 282 students in physical education classes who all followed the same sports program under the same environmental conditions. When evaluating the context and the amount of external sports activities we found no significant difference between the students who developed patellofemoral pain during this

2-year follow-up and the students who did not develop patellofemoral pain. This indicates that both groups were comparable concerning external load on the patellofemoral joint and implies that the incidence of patellofemoral pain in this study was mainly a result of intrinsic risk factors. Because of the small number of subjects with patellofemoral pain in this study, we were unable to statistically compare both groups concerning the type of external sports activity.

The results of our study are in agreement with the findings of Milgrom et al.⁴² in that they indicate that from a broad variety of parameters only a few contribute significantly to the development of patellofemoral pain. The work of Milgrom et al. revealed that the presence of genu varum and a high isometric strength of the quadriceps muscles had a significant correlation with the incidence of patellofemoral pain. They concluded from these data that these parameters were risk factors for this syndrome. In contrast to their study, our results indicate that a decreased flexibility of the quadriceps muscles, a shortened reflex response time of the vastus medialis obliquus muscle, a reduction of the vertical jump performance, and an increase in medial patellar mobility play a significant role in the development of the patellofemoral pain syndrome. Looking at both studies, at first we observe a difference in measurements.

In regard to the medial tibial intercondylar distance, Milgrom et al. measured this with the recruit standing, while we measured it with the student supine. Both methods are, in our opinion, valuable, but they might give different information. The standing measurement probably gives more functionally interesting information, but it is more affected by tibial, ankle, and foot alignment in comparison with the supine measurement. Therefore, the differences in methods may partly account for the differences between studies. Furthermore, the four parameters identified as predisposing factors of patellofemoral pain in our study (a shortened quadriceps muscle, a hypermobile patella, an altered vastus medialis obliquus muscle reflex response time, and decreased explosive strength) were not measured in the study of Milgrom et al.

Moreover, there seems to be a difference between studies concerning the patient selection and the subject's exposure to activity. Milgrom et al. mention that the basic training of the recruits in their study was particularly demanding and progressive in nature. Likely, the less vigorous activity among the subjects in our study contributes to the extremely small number of subjects who developed patellofemoral pain, which limits the interpretation and results.

As mentioned before, the criteria for diagnosing the patellofemoral pain syndrome remain controversial. In addition, the difficulty of differentiating the cause of this pathologic syndrome may, in many cases, contribute to the discrepancy between studies. Milgrom et al. did not give a detailed description of the criteria used for diagnosing patellofemoral pain syndrome. They mention only that recruits who had both subjective and objective findings in either patellofemoral joint were considered to have the syndrome. Therefore, a difference in patient selection be-

tween studies may have occurred, which may partly elucidate the difference in results.

Very conspicuous in this present study is the finding that statistical analyses did not identify any lower leg alignment characteristics nor isokinetic strength as a predisposing factor of patellofemoral pain. This suggests that these parameters may not play a significant role in the development of patellofemoral pain syndrome in an athletic population, in contrast to what is frequently stated. However, since we evaluated a very homogeneous group of young athletes and had a rather small number of patients with patellofemoral pain, one must be very careful when applying these results to the general population of patients with patellofemoral pain.

The finding of decreased quadriceps muscle flexibility in our subjects with patellofemoral pain is in agreement with previous retrospective studies.^{16,32,51} However, since our data were obtained from a prospective study we are able to state that decreased quadriceps muscle flexibility is not necessarily a result of patellofemoral pain syndrome; it existed in these subjects before the pathologic syndrome occurred. These results support the concept that tight quadriceps muscles create high patellofemoral stresses during sports or daily life activities, and in this way predispose patients to patellofemoral pain syndrome. This finding implies that a thorough stretching program would have a beneficial effect in the prevention and the treatment of patellofemoral pain in an adolescent athletic population. However, scientific information about the impact of a thorough stretching program on the functional outcome of patients with patellofemoral pain is, as far as we know, not available in the literature. We therefore believe that future research should be undertaken with regard to the effectiveness of correcting the tightness of the quadriceps and gastrocnemius muscles. This would also give us more insight into the role of muscular tightness in patellofemoral pain syndrome.

Strengthening of the vastus medialis obliquus muscle is considered one of the essential components of the nonoperative approach to patellofemoral pain.^{16,34,38,40,43,52,53} Recent studies have concluded that not only the strength of this muscle but also the time to activation is disturbed in patients with patellofemoral pain.^{60,65} These studies have also shown that the activation pattern of the vastus medialis obliquus and vastus lateralis muscles is disturbed in patients with patellofemoral pain. The authors found a significantly shorter reflex response time for both muscles in patients with patellofemoral pain compared with a control group. In addition, the patients in these studies showed a reversal of the normal firing pattern of the vastus medialis obliquus and vastus lateralis muscles. Whereas in normal subjects the vastus medialis obliquus muscle had a significantly shorter reflex response time compared with the vastus lateralis muscle, the patellofemoral pain patients showed the fastest response time to a patellar tendon tap in the vastus lateralis muscle. In contrast to these findings, one study did not observe an alteration in reflex response time in patients with patellofemoral pain.³⁰ These differences in reflex response times in patellofemoral pain patients between the studies

may be related to methodologic differences.^{29,64} Yet, the results of the reflex response times were all obtained from retrospective studies. On the basis of these investigations, the question remains whether these reflex response times can be predictive for the development of patellofemoral pain syndrome.

In this prospective study using logistic regression analysis, we showed a significant correlation between the presence of patellofemoral pain and an altered vastus medialis obliquus muscle reflex response time. This result is important to the clinician because it implies that we should give great attention to the restoration of the activation pattern of the vastus medialis obliquus muscle. This result also gives scientific support to the goals of several nonoperative treatment programs where attention is focused on normalization of the firing pattern of the vastus medialis obliquus muscle.^{40,46} Recently, Gilleard et al.¹⁷ proved that the application of tape might be helpful in the treatment of this neuromotor disorder. Nonetheless, in the long run, we believe only the use of proper exercises could have a favorable effect on the activation patterns of these muscles. Since no scientific evidence exists on whether exercises can alter these muscle activation patterns, further research on this topic is required. In addition, it remains unclear whether a change in this muscle activation pattern (if possible) would have any effect on the functional outcome of these patients. Further research is needed to clarify this issue.

If the goal of a nonoperative treatment program is to change the muscle activation pattern of the vastus medialis obliquus and vastus lateralis muscles, treatment may consist of three stages: formal motor skill training, incorporation of the motor skill into light functional tasks, and progression to heavy functional tasks.⁴⁸ To allow patients to pass through these different stages of motor relearning more easily, we suggest the use of biofeedback aids. Studies in the movement science and neurophysiology literature prove that feedback is essential to all coordinated movement, and to learning new movement patterns.^{1,19,31}

Besides a decreased flexibility of the quadriceps muscle and a shortened reflex response time of the vastus medialis obliquus muscle, this study also identified a decreased explosive strength capacity as an intrinsic risk factor for patellofemoral pain. A decline in strength of the quadriceps and hamstring muscles has frequently been cited as an etiologic factor of patellofemoral pain.^{2,3,10,11,32,42} However, the present study did not discriminate between the two groups on isokinetic strength parameters. These data suggest that a strength deficit may play an important role in the development of patellofemoral pain, but only if there is a loss of functional explosive strength capacity. We hypothesize that a decreased explosive strength leads to a reduced muscular capacity to absorb high patellofemoral forces during fast functional (eccentric) sports activities. This reduction in capacity of shock absorption may lead to higher patellofemoral stresses, which predispose a person to patellofemoral pain problems. The isokinetic variables could be less valid in measuring this absorbing capacity since in this study they were measured only concentrically and at lower than functional speeds.

Our results are in agreement with the retrospective findings of Thomée et al.,⁵⁶ who reported a decrease in vertical jump performance in patients with patellofemoral pain compared with a control group. Since it is well documented that a good training protocol can significantly increase explosive strength capacity, we strongly recommend this type of training in prevention and treatment programs.

Finally, we found mediolateral patellar mobility was a significant discriminator between the injured and the uninjured groups. Other authors have considered excessive medial and lateral patellar movement clinically significant in the development of the patellofemoral pain.^{14, 15, 33, 47} Biomechanical studies reveal that the patella undergoes a rapid medial translation when it descends into the intercondylar notch. Fairbank et al.¹⁴ state that it is possible that an increased medial laxity of the patella causes repeated blows by the medial facet on the patella against the femoral condyle, which could be responsible for the development of patellofemoral pain.

Because the measurement of mediolateral patellar mobility in this study was performed in subjects with a relaxed quadriceps muscle, the retinaculum may be considered as the most important restraint for this movement. Since it has been proven that exercise strengthens the ligaments by increasing their collagen content,^{57, 58} it is possible that a rehabilitation program could decrease the mediolateral patellar mobility. This might play an important role in the yet unexplained observed pain reduction after a rehabilitation program.^{7, 13, 25}

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